

JavaScript BigInt Datatype - Notes

1. What is BigInt?

BigInt is used to store very large integers beyond the safe limit of normal number datatype.

2. Syntax

```
let a = 12345678901234567890n;  
let b = BigInt(100);
```

3. Typeof

```
typeof 100n -> bigint
```

4. Why Needed?

Normal numbers lose precision after 9007199254740991.
BigInt stores large integers accurately.

5. Arithmetic

```
10n + 20n -> 30n  
10n - 5n -> 5n  
10n * 2n -> 20n  
10n / 3n -> 3n
```

6. Important Rule

Do not mix number and BigInt directly.
10n + 5 -> Error
10n + 5n -> Correct

7. Comparison

```
10n > 5n -> true  
10n == 10 -> true  
10n === 10 -> false
```

8. Boolean Conversion

```
Boolean(1n) -> true  
Boolean(0n) -> false
```

9. Real Uses

Large IDs, financial values, scientific counters, huge population counts.

10. Best Practice

Use BigInt only when large integers are required.

11. Number vs BigInt

Feature	number	bigint
Decimals	Yes	No
Very Large Integers	Limited	Excellent
typeof	number	bigint
Example	100	100n